BEATRICE BAAH

Software & AI Engineer · Data & Security Background
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PROFILE

Software engineer who builds full-stack web and mobile applications and AI-powered products with React, React Native, TypeScript, Node.js, Python and Firebase, integrating Claude, Gemini and OpenAI APIs (including retrieval-augmented search). Came to software from five years of statistics and data-quality work in Ghana's local government and national statistical service, plus an MSc in Cybersecurity and Digital Forensics, and builds with the same priorities: correct data, secure handling of personal records, and products that real users adopt.

TECHNICAL SKILLS

Languages & frameworks: TypeScript, JavaScript, React, React Native (Expo), Node.js/Express, Python, FastAPI, Astro, Flutter (intermediate), SQL

Cloud & data: Firebase (Auth, Firestore, Cloud Functions, Hosting), Google Cloud Run, PostgreSQL, MongoDB, Power BI, STATA, Excel

AI integration: Claude, Gemini and OpenAI APIs; embeddings and vector search (RAG); CNN image classification (TensorFlow/Keras); AI-assisted development with Claude Code

Security: Detections mapped to MITRE ATT&CK, network traffic analysis, anomaly detection (Isolation Forest), SIEM (Splunk, Wazuh), threat-intelligence APIs, CSP and security headers, adversarial ML, Linux and Bash automation

SELECTED PROJECTS

Amansie Ride: Ride-Hailing Platform — React Native · Expo · Firebase · Cloud Functions · React

- Ride-hailing for the districts the big apps don't serve: a rider app, a driver app and an operations console on one backend, with live dispatch, the full trip lifecycle, cash and Mobile Money records, zone-based fares and ratings, all on Firestore listeners and Cloud Functions. In pilot; private repository.

ProAssess: AI Hiring Assessment Platform — React 19 · TypeScript · Firebase · Gemini

- Recruiters invite candidates by email; candidates take four timed reasoning tests; Gemini writes a score summary on submission. Built the candidate pipeline, a side-by-side shortlisting view and GDPR-compliant deletion of candidate data. github.com/Beatbaah/ProAssess

mySOC: Network Security Monitoring & Incident Response Console — Python · FastAPI · PostgreSQL · React

- A sensor turns packet capture and device logs into flows, runs detections mapped to MITRE ATT&CK plus Isolation Forest anomaly scoring, and enriches them with threat intelligence (AbuseIPDB, AlienVault, VirusTotal) into a deduplicated, risk-scored alert queue. Analysts triage, open cases, hunt and hand over between shifts. github.com/Beatbaah/mySOC

Bekwai Municipal Assembly Website Redesign — Astro · TypeScript · Static · Security headers

- Replaced an end-of-life Drupal site served over plain HTTP with fast static pages: HTTPS-only with a strict Content Security Policy, about 75 KB to first view (down from 3.7 MB), and organised around what residents come to do.

Bekwai Tax Revenue: Municipal Revenue Management System — React · Firebase · Cloud Functions

- Designed and built, alone, a revenue system for the Bekwai Municipal Assembly: revenue collection, taxpayer database, electronic payment, receipt verification, and a dashboard of collected and outstanding revenue.
- Repository kept private because it holds taxpayers' personal and financial records; walkthrough available on request.

Dream Trip Planner: AI Travel Agent — Flutter · Node.js · Claude · OpenAI · Cloud Run

- Users describe a trip by voice, text or photos; Claude interprets the request, OpenAI embeddings retrieve similar trips from a vector database, and Claude writes a personalised summary per match. github.com/Beatbaah/dream-trip-planner

CivicMind AI: Municipal Intelligence Platform — React 19 · TypeScript · MongoDB · Gemini

- Tracks municipal revenue and department performance; Gemini turns figures into readable insight, with a rules-based fallback when no API key is available. github.com/Beatbaah/civicmind-ai

Excellent Susu: Savings Group Management App — React · Firebase

- Mobile-first app for rotating savings groups with four role levels, contributions, payouts, reminders, receipts and an audit trail. github.com/Beatbaah/susu-management-app

Client websites (live) Firebase · Firestore

- Corporate World: office-wear e-commerce shop with authentication, cart, wishlist and checkout; Firestore security rules make orders immutable once placed (corporate-world-gh.web.app).
- Haske Spa & Events: booking site for a spa and event business, with a booking flow that routes into WhatsApp (haskespaandevents.web.app).

PROFESSIONAL EXPERIENCE

Municipal Statistician; Acting Head, Statistics Department Nov 2023 – Present

Local Government Service, Bekwai Municipal Assembly, Ghana

- Compile and validate administrative data from all Assembly departments each quarter and report to the Ghana Statistical Service.
- Built Power BI dashboards used by 50+ stakeholders, reducing decision-making delays by about 40%.
- Prepare revenue trend analyses and forecasts for the Municipal Coordinating Director, and figures for the development plan and budget; supervise and train department staff.

Constituency Collation Officer (2024 General Election) Nov – Dec 2024

Electoral Commission of Ghana, Bekwai Constituency

- Collated and verified presidential and parliamentary results from all polling stations, cross-checking result sheets for consistency.

Zonal Data Quality Monitor Dec 2023 – Apr 2024

Ghana Statistical Service, Bekwai / Amansie West

- Supervised data quality for 188 field staff across 288 enumeration zones in the Integrated Business Establishment Survey, using STATA and Excel checks to catch and correct errors.

Assistant Statistician Jan 2022 – Aug 2023

Local Government Service, Bekwai Municipal Assembly

- Analysed municipal datasets and produced Excel and Power BI reports for planning and policy.

Data Quality Monitor (2021 Population and Housing Census) Jun – Sep 2021

Ghana Statistical Service, Ga North Municipal

- Used STATA daily to find and resolve data errors across 186 enumeration areas.

Data Analyst (National Service) Sep 2019 – Sep 2020

Social Security and National Insurance Trust (SSNIT), Kumasi

- Validated and cleaned contribution data and prepared contribution reports for client companies.

EDUCATION

MSc Cybersecurity and Digital Forensics 2024 – 2026

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

- Thesis: a CNN (TensorFlow/Keras) that classifies Ghana Card-style face photos as clean or carrying an adversarial patch ("spot evasion" attack), with a discussion of how such detectors hold up against attackers who adapt to them.

BSc Statistics, Second Class Honours (Upper Division) 2015 – 2019

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

CERTIFICATIONS

Certified Cybersecurity Educator Professional (CCEP), 2025 · Google Cybersecurity Certificate, 2024 · IBM Respond and Mitigate Security Threats, 2024 · Mastercard Cybersecurity Job Simulation (Forage), 2024 · SQL Fundamentals (Dataquest), 2023 · Python Programming Foundations (Udacity), 2022 · Power BI in Data Analytics (Forage), 2022 · Excel Fundamentals: Formulas for Finance (CFI), 2022